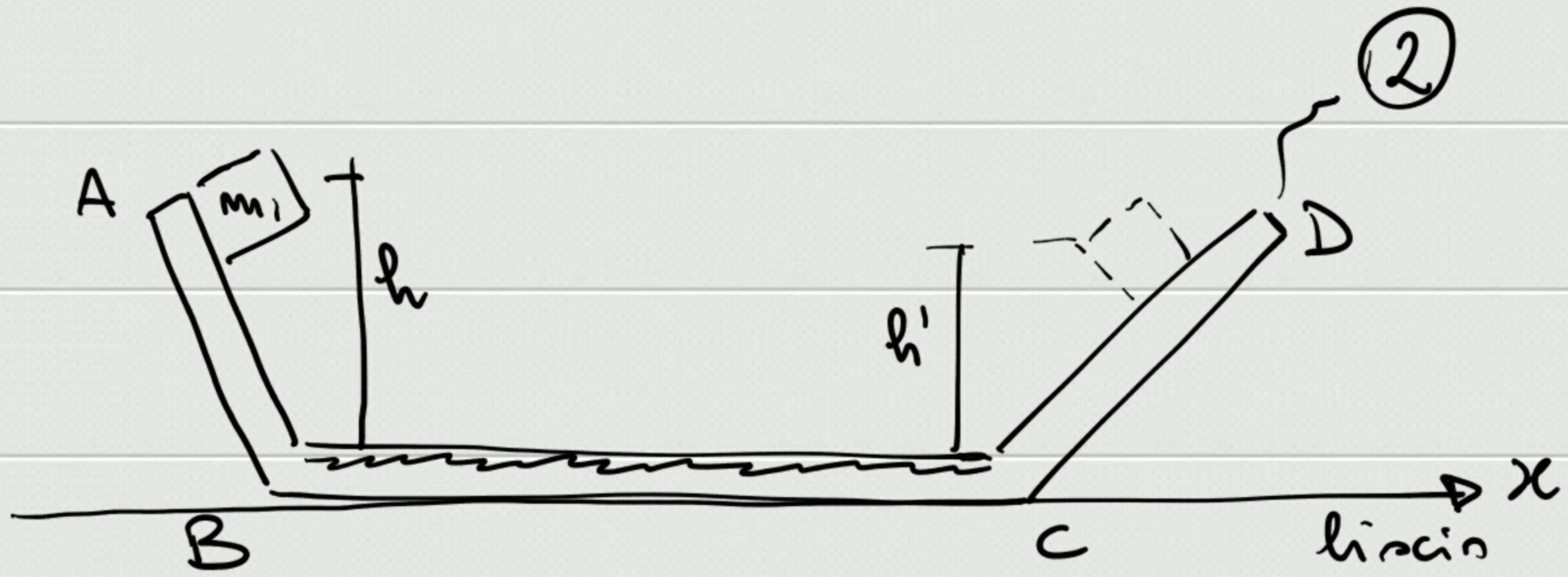


m_2, m_1

$h = 0.3\,m$ $\quad |\overline{BC}| = l = 0.8\,m$ $\quad \mu = 0.15$, $v_{1_0} = v_{2_0} = 0$

$v_2(m_1 \to h') = ?$

$\bar{R}^E \neq 0 \Rightarrow \bar{P} \neq cost$

$R^E_x = 0 \Rightarrow \boxed{P_x = cost} \quad \boxed{m_1 v_{1x} + m_2 v_{2x} = 0}$

$\overset{\shortparallel}{0}$

$$\boxed{v_{1x} = -\frac{m_2}{m_1} v_{2x}} \qquad *$$

$\bar{v} = \bar{v}' + \bar{v}_{0'} \Rightarrow \bar{v}_1 = \bar{v}_1' + \bar{v}_2$

$v'_{1,x} = v_{1,x} - v_{2x}$

$\bar{v}_1'(h') = 0 \Rightarrow v_{1x}(h') = v_{2x}(h')$

$* \Rightarrow v_{1x} = v_{2x} = 0$

$\Rightarrow \boxed{v_2(m_1 \to h') = 0}$

$h' = ?$

$\cancel{h' = h}$

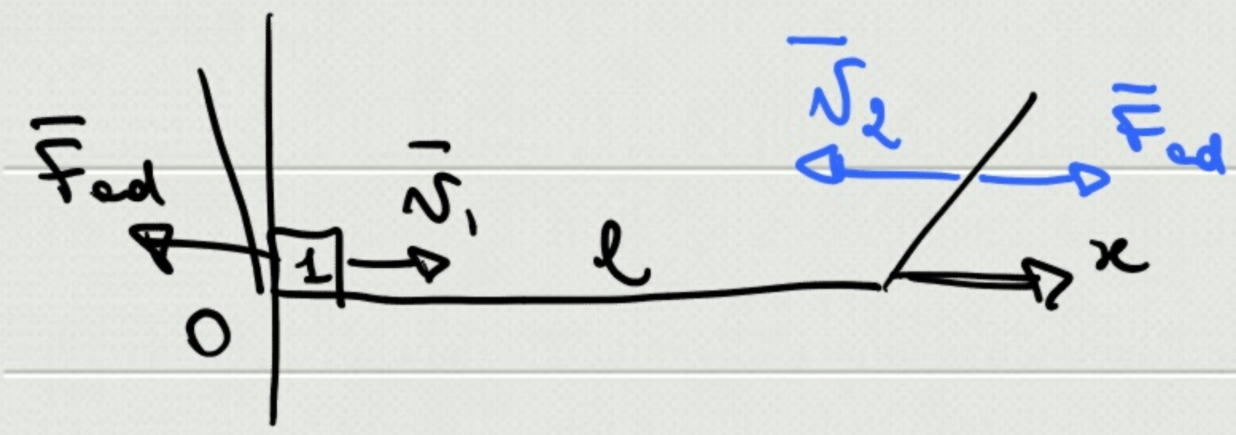
$\mu m_1 g \ell = m_1 g (h' - h)$ **NO**

$-\mu m_1 g \ell = \frac{1}{2} m_1 v_{1C}^2 - \frac{1}{2} m_1 v_{1B}^2$ **NO**

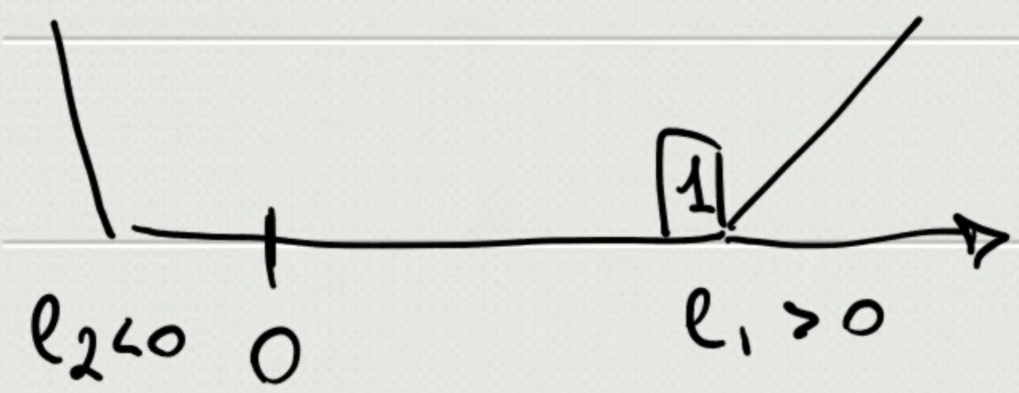
$$\boxed{-\mu m_1 g \ell = m_1 g (h' - h)} \quad *$$

$W_{mc} = \Delta E_m$ $\qquad \Delta E_m = \cancel{\Delta E_k} + \Delta E_p =$

$= m_1 g h' - m_1 g h$



$W_{mc} = W_{mc,1} + W_{mc,2}$



$W_{mc,1} = -\mu m_1 g \ell_1 \quad (<0)$

$W_{mc,2} = -\mu m_1 g \int_0^{\ell_2} \bar{v}_r \, d\bar{s} =$

$= +\mu m_1 g \int_0^{\ell_2} ds =$

$= \mu m_1 g \ell_2 \quad (<0)$

$W_{mc} = -\mu m_1 g \ell_1 + \mu m_1 g \ell_2 =$

$= \mu m_1 g \underbrace{(\ell_2 - \ell_1)}_{=-\ell} = -\mu m_1 g \ell$

$-\mu \cancel{m_1} g \ell = \cancel{m_1} g (h' - h) \Rightarrow h' = h - \mu \ell = 0.18 \text{ m}$

$v_2 \, (v_{1f} = 0) = ?$ $\Rightarrow$ $\boxed{v_{2,f} = 0}$

$d_{attrito} = ?$

$$-\mu \cancel{m_1} g d = 0 - \cancel{m_1} g h \Rightarrow d = \frac{h}{\mu} = 2 \, m$$